Started on	Sunday, 24 July 2022, 4:39 PM
State	Finished
Completed on	Sunday, 24 July 2022, 4:39 PM
Time taken	8 secs
Grade	**0.00** out of 105.00 (**0**%)

Question **1**

Not answered

Marked out of 1.00

Flag question

John Wilson is a 37-year-old Caucasian male with treatment-resistant schizophrenia. He meets with his psychiatrist to discuss the potential side effects of the antipsychotic clozapine. What is the risk of agranulocytosis in a patient taking clozapine after one year?

Select one:

- 0.5%
- 0.01%
- 0.1%
- 5%
- 1%

Your answer is incorrect.

Explanation:

The prevalence of agranulocytosis during clozapine treatment is 0.4%, lower than previously thought. Clozapine-induced agranulocytosis typically occurs within the first 18 weeks of treatment, with few cases occurring beyond 6 months. After 1 year of treatment, residual risk is estimated at 0.39/1000 patient/years (risk of <0.01% after one year).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 236.

Mijovic A, MacCabe JH. Clozapine-induced agranulocytosis. Annals of Hematology. 2020 Aug 20:1-6.

The correct answer is: 0.01%

Question **2**

Not answered

Marked out of 1.00

Flag question

Mr. Davies is a 32-year-old man with a diagnosis of bipolar affective disorder who has required prolonged admissions in the past for episodes of mania. He has now presented to clinic with a severe depressive episode despite being compliant with mood stabilising medication. He is wondering about taking an antidepressant but is worried about the risk of a manic episode and refuses to take olanzapine in combination due to severe weight gain in the past. Which of the following antidepressants is least likely to induce a manic switch while already taking long-term treatment for bipolar disorder?

Select one:

- Vortioxetine

- Sertraline
- Venlafaxine
- Phenelzine
- Paroxetine

Your answer is incorrect.

As reflected in the most recent bipolar guidelines, there are several options, including the addition of olanzapine+fluoxetine, quetiapine, olanzapine, or lamotrigine. But this patient refuses olanzapine.

The meta-analytical literature (Gijsman et al, 2004) provides a rate of switching for tricyclic antidepressants as 10% compared with 3.2% for all other antidepressants combined. MAOIs are also categorised to have a higher switch rate (Yatham, Kusumakar & Kutcher, 2002; Page 24). There is a 4-fold higher switch rate for venlafaxine (29%) compared with SSRIs (sertraline 9%) and bupropion (10%) when used concurrently with a mood stabilizer. Fluvoxamine may have much lower risk and newer agents such as vortioxetine may have a higher switch rate, but further studies are needed to reduce the uncertainty.

Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4321017/ .

Also see Jo YT et al. Diagnostic conversion from unipolar to bipolar affective disorder: a population-based study. Journal of Affective Disorders. 2022 Jan 20.

National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

The correct answer is: Sertraline

Question **3**

Not answered

Marked out of 1.00

Flag question

Sadira is a 36-year-old female from South India. She is an inpatient on your ward and was diagnosed with treatment-refractory schizophrenia five months ago. Since then she has been treated with psychotropic medication, which she initially tolerated well. She says that her auditory hallucinations have lessened in their intensity. Previously a heavy smoker, Sadira managed to give this up recently. During the ward round she complains of feeling increasingly drowsy. You look at her medications to see if she is taking anything that may be contributing to this. Which of the following medications are most likely contributing to her current presentation?

Select one:

- Sertraline
- Clozapine
- Zopiclone
- Paracetamol
- Mirtazapine

Your answer is incorrect.

Explanation:

Tobacco smoke contains polycyclic aromatic hydrocarbons that induce (increase the activity of) certain hepatic enzymes (CYP1A2 in particular). The extent of enzyme induction is determined by the number and type of cigarettes smoked and the degree of smoke inhalation. For some drugs used in psychiatry, smoking significantly reduces drug plasma levels and higher doses are required than in

non-smokers. Smoking reduces plasma clozapine levels by up to 50%. When people stop smoking, enzyme activity halves roughly every 2 days. Plasma levels of affected drugs will then rise, sometimes substantially. Dose reduction will usually be necessary.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 856.

The correct answer is: Clozapine

Question **4**

Not answered

Marked out of 1.00

Flag question

Mr. Collins is a 48-year-old man with a diagnosis of alcohol dependence. He previously had 10 years of abstinence after treatment with disulfiram but relapsed 3 years ago. He is keen to have disulfiram again. Which of the following is a side-effect of disulfiram in the absence of alcohol consumption?

Select one:

- Excessive sweating
- Nausea
- Headache
- Flushing
- Palpitations

Your answer is incorrect.

Explanation:

In the absence of alcohol consumption, side effects of disulfiram include halitosis, metallic taste, headache, dermatitis, and sedation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 598.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 239.

The correct answer is: Headache

Question **5**

Not answered

Marked out of 1.00

Flag question

Mr Edwards is a 60-year-old man who has erectile dysfunction. He attends his GP asking about sildenafil. Which of the following is a common side-effect of this medication?

Select one:

- Diarrhoea
- Nasal congestion
- Nausea
- Polydipsia
- Constipation

Your answer is incorrect.

Explanation:

Common side effects of phosphodiesterase inhibitors (such as sildenafil and tadalafil) are mild

headaches, dizziness, and nasal congestion.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Nasal congestion

Question **6**

Not answered

Marked out of 1.00

Flag question

Mr. Fowler is a 30-year-old man with a diagnosis of bipolar affective disorder. His main difficulty is episodes of depression. He has attended clinic asking about the possibility of starting lamotrigine. Which of the following is a side-effect of this medication?

Select one:

- Constipation
- Diarrhoea
- Sialorrhoea
- Rash
- Nasal congestion

Your answer is incorrect.

Explanation:

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g. Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with a number of systemic manifestations). Rash is most likely to occur within the first 2-8 weeks of treatment and is more likely when combined with valproate, exceeding the recommended initial dose, or rapid dose escalation.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 358.

The correct answer is: Rash

Question **7**

Not answered

Marked out of 1.00

Flag question

Some psychiatric medications can interfere with anticoagulants with potentially severe consequences. Which of the following combinations of medication would indicate a need for an urgent INR?

Select one:

- Unfractionated heparin and fluoxetine
- Warfarin and mirtazapine
- Warfarin and St John's Wort
- Rivaroxaban and olanzapine
- Rivaroxaban and fluoxetine

Your answer is incorrect.

Explanation:

St. John’s wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin

content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: Warfarin and St John's Wort

Question **8**

Not answered

Marked out of 1.00

Flag question

An ECG for your patient Will Hatton (aged 46) shows a QTc of 480m sec. Which of the following medications has the least effect on QTc interval?

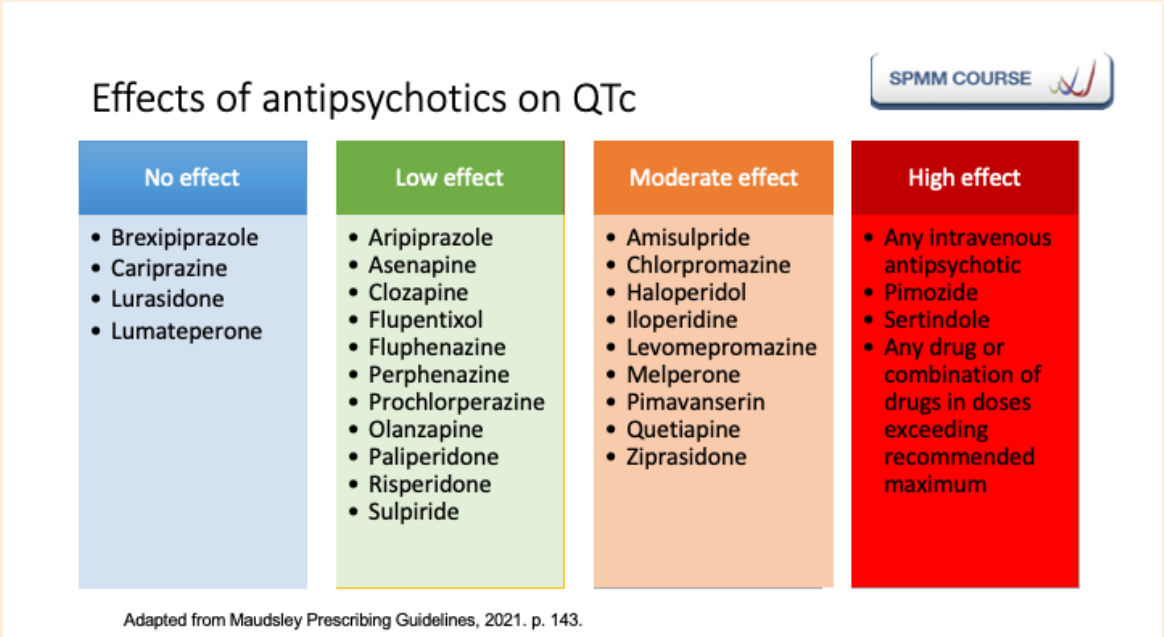
Select one:

- Clomipramine
- Citalopram
- Quetiapine
- Ziprasidone
- Aripiprazole

Your answer is incorrect.

Explanation:

Aripiprazole has the lowest effect on QTc.



Antidepressant	Effect on QTc (adapted from Maudsley Prescribing Guidelines, 2021. pp. 383-5.)
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
Citalopram/escitalopram	Dose-related increase in QTc
Duloxetine	Isolated reports of prolongation
Fluoxetine	No effect on QTc interval
Fluvoxamine	No significant effect on QTc
Levomilnacipran	No effect on QTc interval
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Milnacipran	No effect on QTc
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval at standard doses; small increase at 400mg/day
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 143, 383-385.

The correct answer is: Aripiprazole

Question **9**

Not answered

Marked out of 1.00

Flag question

Mr. Ahmed is a 58-year-old man who has been experiencing depressive symptoms following the loss of his job four months previously. Given he has a strong family history of cardiac dysfunction, the GP arranges for a routine ECG prior to prescribing an antidepressant. There is nothing of concern on the ECG, so the GP prescribes an antidepressant. After 3 months, the patient took an overdose and the ECG now shows QTc prolongation. Of the list given, which antidepressant is most associated with prolonged QTc?

Select one:

- Moclobemide
- Sertraline
- Reboxetine
- Mirtazapine
- Venlafaxine

Your answer is incorrect.

Explanation:

Of the list given, Moclobemide is possibly arrhythmogenic in overdose and QTc prolongation can occur in overdose.

Antidepressant	Effect on QTc (adapted from Maudsley Prescribing Guidelines, 2021. pp. 383-5.)
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
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Fluvoxamine	No significant effect on QTc
Levomilnacipran	No effect on QTc interval
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Milnacipran	No effect on QTc
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval at standard doses; small increase at 400mg/day
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 383-385.

The correct answer is: Moclobemide

Question **10**

Not answered

Marked out of 1.00

Flag question

Dawn is a 34-year-old woman with a diagnosis of epilepsy, who you have seen in clinic because of her struggles with low mood. The neurologist makes a recommendation on adding another medication to her current regime of carbamazepine. You are aware that carbamazepine can raise plasma levels of which one of the following drugs?

Select one:

- Sodium Valproate
- Phenytoin
- Lithium
- Lamotrigine
- Topiramate

Your answer is incorrect.

Explanation:

Carbamazepine can increase plasma levels of clomipramine, phenytoin, and primidone. When carbamazepine is used concurrently, phenytoin plasma levels may increase to toxic levels. Careful titration and monitoring is advised.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 134

The correct answer is: Phenytoin

Question **11**

Not answered

Michael has a history of psychosis. He is currently on a renal dialysis unit as his GFR is less than 10 ml/min. He has established renal failure. Which medication should be avoided in established renal

Marked out of 1.00

Flag question

impairment?

Select one:

- Haloperidol
- Aripiprazole
- Amisulpride
- Olanzapine
- Risperidone

Your answer is incorrect.

Explanation:

Amisulpride and sulpiride are almost totally renally excreted and best avoided in renal impairment and cases with established renal failure.

Drug group	Recommended psychotropics in renal impairment
Antipsychotics	No agent clearly preferred to another, however: **Avoid** sulpiride and amisulpride **Avoid** highly anticholinergic drugs because they can contribute to urinary retention First-generation antipsychotic: suggest **haloperidol** 2-6mg/day Second-generation antipsychotic: suggest **olanzapine** 5mg/day
Antidepressants	No agent clearly preferred to another, however: **Citalopram** (care re: QTc-prolonging effects) and **sertraline** are reasonable choices **CBT** where available
Mood stabilisers	No agent clearly preferred to another, however: **Avoid** lithium if possible Suggest starting **valproate or lamotrigine** at low dose, increase slowly, and monitor for adverse effects
Anxiolytics and hypnotics	No agent clearly preferred to another, however: Excessive sedation more likely to occur, so monitor all patients carefully **Lorazepam** and **zopiclone** are reasonable choices
Anti-dementia drugs	No agent clearly preferred to another, however: Rivastigmine is a reasonable choice

Adapted from Maudsley Prescribing Guidelines 2021. p. 749.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 749.

The correct answer is: Amisulpride

Question **12**

Not answered

Marked out of 1.00

Flag question

Nadine is a 46-year-old French teacher with a diagnosis of bipolar affective disorder. You wish to commence her on lithium and enquire about her medication history to exclude possible interactions. Which of the following medications reduces lithium excretion?

Select one:

- Propanolol
- Ramipril
- Sertraline
- Amiloride
- Lamotrigine

Your answer is incorrect.

Explanation:

ACE inhibitors (such as ramipril) can reduce thirst (which can lead to mild dehydration) and increase renal sodium loss (leading to increased sodium re-absorption by the kidney), resulting in an increase in lithium plasma levels. The magnitude of this effect is variable, from no increase to a four-fold increase. The risk seems to be increased in patients with heart failure, dehydration, and renal impairment. In the elderly, ACE inhibitors increase seven-fold the risk of hospitalisation due to lithium toxicity. ACE inhibitors can also precipitate renal failure so, if co-prescribed with lithium, more frequent monitoring of eGFR and plasma lithium is required.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 252-253.

The correct answer is: Ramipril

Question **13**

Not answered

Marked out of 1.00

Flag question

Mark attends clinic complaining of a side effect of his medication. Which of the following describes a slow rhythmic movement disorder affecting the extremities most markedly?

Select one:

- Dystonia
- Tic
- Sydenham chorea
- Akathisia
- Athetosis

Your answer is incorrect.

Explanation:

Athetosis is characterised by involuntary, slow, writhing movements, especially of the fingers, hands, head, and tongue. Akathisia is a feeling of inner restlessness and desire to move. A tic is a habitual spasmodic contraction of muscles. Sydenham chorea is rapid uncoordinated jerky movements.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Athetosis

Question **14**

Not answered

Marked out of 1.00

Flag question

Miss Clarke is a 32-year-old woman with a diagnosis of opiate dependence syndrome who is under a substance misuse service. She attends clinic and sees you to request that you prescribe her an alpha-2-agonist medication to prevent her experiencing withdrawals whilst she stops using opiates. She has a history of experiencing marked hypotension in the past. Which of the following medications would be most appropriate for Miss Clarke?

Select one:

- Romifidine
- Lofexidine
- Guanfacine

- Clonidine
- Tizanidine

Your answer is incorrect.

Both Clonidine and Lofexidine are used to reduce the withdrawal symptoms associated with stopping opiates. Clonidine is commonly associated with hypotension, whilst this is not seen with lofexidine with the same frequency. Interesting, clonidine can be used in some patients with profound and persistent hypotension due to it having a limited central hypotensive effect in these patients.

The correct answer is: Lofexidine

Question **15**

Not answered

Marked out of 1.00

Flag question

Which of the following best describes how methylphenidate works?

Select one:

- Inhibition of the hypothalamus
- Serotonin reuptake inhibitor
- Blocking dopamine transporter
- Blocking orexin
- Histamine antagonist

Your answer is incorrect.

Explanation:

Stimulants (perhaps better designated as norepinephrine (NE) and dopamine (DA) reuptake blockers) such as methylphenidate block the transporters both for NE and DA (NETs and DATs, respectively). Methylphenidate blocks DATs and NATs allosterically, stopping the reuptake of DA via DATs and NE via NETs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 469.

The correct answer is: Blocking dopamine transporter

Question **16**

Not answered

Marked out of 1.00

Flag question

When she was a teenager, Gemma struggled with her self-esteem and self-image. She exercised several times a day and severely restricted her dietary intake. Despite this, she wanted to lose further weight and bought some pills online. Her GP had also recently increased Gemma's dose of venlafaxine the previous week with the hope of helping Gemma with her feelings of low mood. Three days after taking the pills bought online, Gemma's mother found her collapsed on her bed, sweating profusely, in an altered conscious state. What medication could Gemma have ordered on the internet that may have caused this?

Select one:

- Magnesium
- Orlistat
- Sibutramine
- Saffron extract

- Guarana

Your answer is incorrect.

Sibutramine (formerly sold as Meridia) is a monoamine reuptake inhibitor, and acts as an appetite suppressant. Risks of concomitant use with SSRIs and SNRIs include serotonin syndrome, which can manifest with confusion, altered consciousness, sweating, muscle rigidity, tachycardia, dilated pupils, and required urgent medical intervention. Sibutramine has been withdrawn from the UK and many other countries.

The correct answer is: Sibutramine

Question **17**

Not answered

Marked out of 1.00

Flag question

Charles is a 53-year-old male with a diagnosis of moderate depressive disorder. His psychiatrist prescribed him amitriptyline two months ago. At his follow up appointment, Charles reports an improvement in his mood and sleep but complains of lightheadedness. His psychiatrist measures his sitting-standing blood pressure and notices a considerable drop in his systolic blood pressure, which satisfies the criteria for orthostatic hypotension. Which receptor blockade leads to this kind of effect?

Select one:

- 5-HT2A receptors
- Alpha-1 receptors
- Histamine
- Beta receptors
- D2 receptors

Your answer is incorrect.

Explanation:

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Its blockade of alpha adrenergic 1 receptor can cause dizziness, sedation, and hypotension. Sedative effects and weight gain can occur due to antihistaminic properties. Anticholinergic activity can cause sedative effects, dry mouth, constipation, and blurred vision.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 25-26.

The correct answer is: Alpha-1 receptors

Question **18**

Not answered

Marked out of 1.00

Flag question

Which of the following describes the mechanism of action of Pregabalin?

Select one:

- Binds GABA-A sites
- Binds to voltage-gated calcium channels
- Potentiates effects of GABA
- Has a mechanism of action identical to GABA
- Blocks glutamate

Your answer is incorrect.

Explanation:

Pregabalin is a leucine analog and is transported both into the blood from the gut and also across the blood-brain barrier into the brain by the system L transport system (a sodium-dependent transporter) as well as by additional sodium-dependent amino acid transporter systems. Pregabalin binds to the alpha 2 delta subunit of voltage-sensitive calcium channels. This closes N and P/Q presynaptic calcium channels, diminishing excessive neuronal activity and neurotransmitter release. Although structurally related to gamma-aminobutyric acid (GABA), it has no known direct actions on GABA or its receptors.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 647

The correct answer is: Binds to voltage-gated calcium channels

Question **19**

Not answered

Marked out of 1.00

Flag question

Blocking dopamine in which of the following pathways would result in hyperprolactinaemia?

Select one:

- Cerebellar
- Mesolimbic
- Nigrostriatal
- Mesocortical
- Tuberoinfundibular

Your answer is incorrect.

Explanation:

The dopamine neurons that project from hypothalamus to anterior pituitary gland are known as the **tuberoinfundibular** dopamine pathway. Normally, these neurons are tonically active and inhibit prolactin release. In the postpartum state, however, the activity of these dopamine neurons is decreased. Prolactin levels can therefore rise during breast feeding so that lactation will occur. If the functioning of the tuberoinfundibular dopamine neurons is disrupted by lesions or drugs, prolactin levels can also rise.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 85.

The correct answer is: Tuberoinfundibular

Question **20**

Not answered

Marked out of 1.00

Flag question

Zopiclone acts on which receptor site?

Select one:

- Delta-1 subunit of Omega 1 receptor
- Omega-1 subunit of Alpha 1 receptor
- Omega-1 subunit of Sigma 1 receptor
- Alpha-1 subunit of Omega 1 receptor
-

Alpha-2 subunit of Omega 2 receptor

Your answer is incorrect.

Explanation:

Z drugs bind to benzodiazepine-sensitive GABA-A receptors. A different nomenclature (omega system) denotes the specific sites of GABA-A receptors where these 'Z' drugs act as omega-1 receptors. Conventional benzodiazepines interact at both ω1 and ω2 sites, the latter likely contributing to side effects.

Zolpidem and zaleplon bind selectively to alpha-1 subunits of these sites. By contrast, benzodiazepines and zopiclone/eszopiclone bind to four alpha subunits (alpha-1, alpha-2, alpha-3, and alpha-5). The functional significance of alpha-1 selectivity is not yet proven, but may contribute to lower risk of tolerance and dependence. The alpha-1 subtype is known to be critical for producing sedation and thus is targeted by every effective GABA-A PAM (positive allosteric modulator) hypnotic, both benzodiazepines and Z-drugs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 422-423.

Madari S, Golebiowski R, Mansukhani MP, Kolla BP. Pharmacological management of insomnia. Neurotherapeutics. 2021 Jan;18(1):44-52.

The correct answer is: Alpha-1 subunit of Omega 1 receptor

Question **21**

Not answered

Marked out of 1.00

Flag question

Mr. Zane is a 60-year-old man who is on warfarin following a DVT. Which of the following medications would be the most important for him to avoid?

Select one:

- Lamotrigine
- Fluoxetine
- Lithium
- Risperidone
- St John's Wort

Your answer is incorrect.

Explanation:

St. John's wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: St John's Wort

Question **22**

Women with mental health conditions often need to continue medication during pregnancy and

Not answered

Marked out of 1.00

Flag question

breastfeeding. The benefit from medicines during this time must be balanced against possible risks to the newborn and neonate. Which of the following drugs is considered to be the most teratogenic?

Select one:

- Dosulepin
- Lithium
- Carbamazepine
- Valproate
- Lamotrigine

Your answer is incorrect.

Explanation:

Valproate is associated with strong evidence for increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language), low IQ, and autism. This risk increases with increasing doses of valproic acid, especially when the dose is more than 1gm/day.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

National Institute for Health and Clinical Excellence. Antenatal and postnatal mental health: clinical management and service guide. Updated edition 2020.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 690-691.

The correct answer is: Valproate

Question **23**

Not answered

Marked out of 1.00

Flag question

Miss Davies is a 28-year-old woman with a diagnosis of opiate dependence. She is keen to become abstinent and asks if there is any medication she could take that would counteract the effect of opioids and help her stop their use. Which of the following is a medication that would best fit this description?

Select one:

- Methadone
- Clonidine
- Naloxone
- Naltrexone
- Baclofen

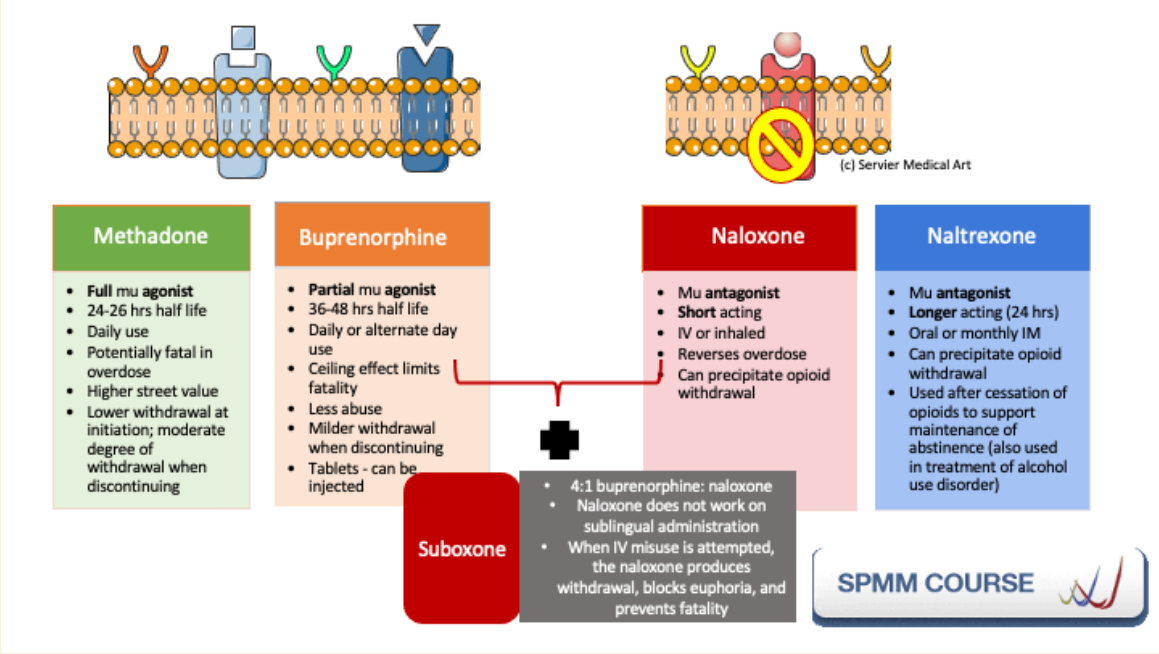
Your answer is incorrect.

Explanation:

Naltrexone has been found by NICE to be a cost-effective treatment strategy in aiding abstinence from opioid misuse for those who prefer an **abstinence programme** (rather than substitute prescribing of methadone or buprenorphine), are **fully informed** of potential risks and benefits, are **highly**

motivated to remain on treatment, and have a **partner supporting concordance.** It hasthe propensity to cause a **severe withdrawal reaction** from opioids. A **test dose of naloxone**, which has a much **shorter half-life** than naltrexone, may be given prior to starting naltrexone treatment; any withdrawal symptoms precipitated will be of shorter duration than if precipitated by naltrexone. **Discontinuation** may be associated with an increase in **inadvertent overdose** from illicit opioids, making **supervision** of administration and **careful choice** of who is prescribed it (**abstinence-focused and motivated**) very important. People taking naltrexone often experience **adverse effects** of unease (dysphoria), depression and insomnia, which can lead to relapse to **illicit opioid use** or **failure to continue treatment.**

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 496.



The correct answer is: Naltrexone

Question **24**

Not answered

Marked out of 1.00

Flag question

Which of the following neurotransmitters is primarily released from the medial and dorsal raphe nucleus?

Select one:

- Acetylcholine
- GABA
- Serotonin
- Dopamine
- Noradrenaline

Your answer is incorrect.

Explanation:

Serotonin has both ascending and descending projections. Ascending serotonergic projections originate mainly in the raphe nucleus in the brainstem. These ascending projections may regulate mood, anxiety, sleep, and other functions. Descending serotonergic projections extend down the brainstem and through the spinal cord. They may regulate pain.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 279.

The correct answer is: Serotonin

Question **25**

Not answered

Marked out of 1.00

Flag question

Rosalind is a 75-year-old female who presents to your clinic with a five-week history of low mood, anhedonia, poor appetite and insomnia. You diagnose her with moderate depressive disorder and wish to start her on antidepressants. You ask her about other medications she is taking and she tells you that she is taking amlodipine and warfarin. Which of the following antidepressants will you avoid prescribing?

Select one:

- Trazadone
- Duloxetine
- Fluoxetine
- Venlafaxine
- Mirtazapine

Your answer is incorrect.

Explanation:

Fluoxetine may displace highly protein-bound drugs like warfarin. Fluoxetine carries a possible increased risk of bleeding, especially when combined with anticoagulants (e.g. warfarin, NSAIDs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 304

The correct answer is: Fluoxetine

Question **26**

Not answered

Marked out of 1.00

Flag question

Miss Miller is a 28-year-old woman who attends her GP describing episodes of irresistible sleep and collapsing when laughing. Which of the following is most associated with this diagnosis?

Select one:

- Loss of orexin-secreting neurons
- Deficiency of leptin peptides
- Histamine deficiency
- Excess melatonin
- Circadian rhythm disturbance

Your answer is incorrect.

The diagnosis is Narcolepsy. A deficiency is orexin peptides, usually caused by a loss of orexin-secreting neurons, is the primary cause of narcolepsy. This can be associated with polymorphisms in the genes coding for HLA proteins.

The correct answer is: Loss of orexin-secreting neurons

Question **27**

Not answered

Marked out of 1.00

Flag question

Which of the following depot injections has the longest acting half-life after the first dose?

Select one:

- Paliperidone Palmitate
- Olanzapine Pamoate Monohydrate
- Fluphenazine Decanoate
- Haloperidol decanoate
- Zuclopenthixol Decanoate

Your answer is incorrect.

The half-life of Paliperidone Palmitate can be upto 40 days (ranges 25 – 49 days). The half-life of the others listed: Olanzapine Pamoate Monohydrate – 30 days, Zuclopenthixol decanoate– 19 days and Haloperidol decanoate– 21 days, Fluphenazine decanoate 6-9 days.

The correct answer is: Paliperidone Palmitate

Question **28**

Not answered

Marked out of 1.00

Flag question

Meghan is a 43-year-old female with paranoid schizophrenia. She is being treated with olanzapine. Her most recent blood test revealed hyponatraemia. Following review by the medical team, she is found to have SIADH (Syndrome of inappropriate ADH secretion). The medics request that her antipsychotic is changed in order to prevent the hyponatraemia from worsening. Which of the following is the most appropriate antipsychotic to use in a patient with hyponatraemia?

Select one:

- Haloperidol
- Chlorpromazine
- Flupentixol
- Quetiapine
- Clozapine

Your answer is incorrect.

Although research is limited, clozapine has been found to be the most effective antipsychotic for schizophrenia in patients with associated hyponatraemia. The Canuso & Goldman study (1999) found that clozapine increased plasma osmolality significantly, thus preventing hyponatraemia and the need for fluid restriction.

The correct answer is: Clozapine

Question **29**

Not answered

Marked out of 1.00

Flag question

Mrs Bells is an 81 year old woman who was brought to the A&E department by her daughter because she was reportedly getting more confused, and feeling weak (not able to stand up from her chair without support), and had begun taking her clothes off seemingly without purpose. You review the history taken by the A&E doctors and note that three days previously Mrs Bells had been started on an antidepressant by her GP. The blood tests show she has a sodium of 129 mmol/L. Of the following, which antidepressant is least likely to be associated with hyponatremia?

Select one:

- Trazodone
- Fluoxetine
- Venlafaxine
- Amitriptyline
- Sertraline

Your answer is incorrect.

There's a moderate risk of hyponatremia with TCAs, but the risk is higher for Trazodone (SARI), and even higher for SSRIs and SNRIs (de Picker et al, Psychosomatics 2014; 55)

The correct answer is: Amitriptyline

Question **30**

Not answered

Marked out of 1.00

Flag question

A medical student in your clinic is discussing the pharmacology of antipsychotics and asks which antipsychotic is a pure D2 antagonist?

Select one:

- Aripiprazole
- Amisulpride
- Perospirone
- Sulpiride
- Olanzapine

Your answer is incorrect.

Sulpiride is a pure D2 antagonist. The other antipsychotics also have high affinity for other receptors including the 5HT receptors.

The correct answer is: Sulpiride

Question **31**

Not answered

Marked out of 1.00

Flag question

Marian attends clinic stating that she wants to change her antipsychotic medication as she is putting on weight. Which of the following is the most weight-neutral antipsychotic?

Select one:

- Olanzapine
- Aripiprazole
- Risperidone
- Quetiapine
- Clozapine

Your answer is incorrect.

Aripiprazole is not associated with weight gain, whereas the other atypical antipsychotics are associated with weight gain.

The correct answer is: Aripiprazole

Question **32**

Not answered

Marked out of 1.00

Flag question

Robin is a 53-year-old male with a history of bipolar affective disorder. He attends your outpatient clinic complaining of low mood, poor appetite, weight loss and hopelessness. Your impression is that he is suffering from a depressive relapse. In the past, his depressive symptoms have been managed with antidepressants; however, quite often this has resulted in the presentation of mania. Which of the following antidepressants are less likely to result in a switch to mania?

Select one:

- Bupropion
- Mirtazapine
- Venlafaxine
- Imipramine
- Amitriptyline

Your answer is incorrect.

A switch to mania is a potential risk of antidepressant therapy in bipolar affective disorder. Bupropion is less likely to induce mania compared with other antidepressant medications provided the dose does not exceed the recommended daily dose of 450mg. Doses higher than this should be used with caution in patients with bipolar affective disorder (Goren & Levin, 2000).

The correct answer is: Bupropion

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following antidepressants is a potent enzyme inhibitor?

Select one:

- Mirtazapine
- Fluoxetine
- Venlafaxine
- Citalopram
- Imipramine

Your answer is incorrect.

Fluoxetine and its metabolite, norfluoxetine, are both potent inhibitors of cytochrome P450 2D6 (CYP2D6) which metabolises them.

The correct answer is: Fluoxetine

Question **34**

Not answered

Tan is a 78-year-old Vietnamese gentleman recently diagnosed with agitated depression. You wish to prescribe medication to manage his condition and have a discussion with the team pharmacist about

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Flag question

the appropriateness of prescribing antipsychotic medication. Which of the following pharmacokinetic changes is most likely to occur in elderly patients taking lipophilic medications?

Select one:

- Increased creatinine clearance
- 25% reduction of GFR by the age of 80
- Increase body water
- Increase volume of distribution
- Lower absorption from GIT

Your answer is incorrect.

In the elderly, an increase in total body fat and a decrease in total muscle mass and total body water results in a larger volume of distribution and therefore longer half-life of lipophilic medications because of their increased sequestration in fat. Absorption is not greatly affected. Decreases in renal blood flow are approximately 10% per decade after the fourth decade, which results in reduced creatinine clearance and GFR.

The correct answer is: Increase volume of distribution

Question **35**

Not answered

Marked out of 1.00

Flag question

Mr. Adams is a 72-year-old man who was brought to his GP by his family as they were concerned about a progressive decline in his memory and ability to recall words. Whilst there his GP noted that Mr. Adams' arms appeared to be stiff. Which of the following imaging modalities would be best to help determine his diagnosis?

Select one:

- DAT scan
- CT scan
- Structural MRI
- PET scan
- Functional MRI

Your answer is incorrect.

Mr Adams is displaying symptoms that could be consistent with Alzheimer's disease or dementia with Lewy bodies. The stiffness in his arms could be consistent with the latter diagnosis. A DAT scan is superior to the others listed as it can demonstrate the specific loss of dopaminergic neurons in the basal ganglia. The other scans listed cannot provide this. Even if they showed a degree of hippocampal volume loss, this would not necessarily preclude a diagnosis of dementia with Lewy bodies. A diagnosis of dementia with Lewy bodies is primarily clinical, although a DAT scan is useful in aiding this.

The correct answer is: DAT scan

Question **36**

Not answered

Mr. Ellison is a 70-year-old man who was taken to his GP with increasing symptoms of cognitive impairment, resting tremor, limb stiffness and episodes when he appears to be seeing distressing

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Flag question

things. Which of the following is the gene product most likely to be associated with his diagnosis?

Select one:

- Alpha-synuclein
- Tau protein
- Amyloid A protein
- Beta-Synuclein
- Ubiquitin

Your answer is incorrect.

Explanation:

The diagnosis is Lewy body dementia (LBD) and the pathological finding in this condition is the presence of Lewy bodies. The key constituent of Lewy bodies is alpha-synuclein. The function of alpha-synuclein is not certain, but it is thought to function in cell membrane remodeling at neuronal terminals. When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons or surrounding glial cells, Lewy bodies are formed.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

The correct answer is: Alpha-synuclein

Question **37**

Not answered

Marked out of 1.00

Flag question

Abdul is a 10-year-old biy who suffers from ADHD. You meet with his parents in your outpatient clinic. They have been reading about Modafinil on the Internet and are keen to learn more. Through which neurotransmitter does modafinil predominantly exert its effect in ADHD?

Select one:

- Serotonin
- GABA
- Acetylcholine
- Dopamine
- Glutamate

Your answer is incorrect.

Explanation:

How modafinil works is unknown, but it is clearly different from classical stimulants such as methylphenidate and amphetamine. Modafinil binds to and requires the presence of the dopamine transporter; it also requires the presence of alpha-adrenergic receptors. It hypothetically acts as an inhibitor of the dopamine transporter.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 521

The correct answer is: Dopamine

Question **38**

Not answered

The movement of ions is an essential part of the transmission of impulses within the nervous system. Which of the following ions flows into neurons to propagate action potentials?

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Flag question

Select one:

- Chloride
- Calcium
- Phosphate
- Potassium
- Sodium

Your answer is incorrect.

Explanation:

An action potential is the momentary change in the voltage difference across the membrane of a neuron, usually triggered by nerve impulses from several other neurons at synapses. It is characterised by an influx of positively charged sodium ions through the cell membrane, resulting in an impulse being propagated along the neuron's axon, reversing its resting potential of approximately -70 millivolts (mV), the inner surface being negative relative to the outer surface and the membrane more permeable to potassium ions than to sodium ions, and changing it to approximately +40 mV, increasing the permeability of the membrane to sodium ions and resulting in the influx of positively charged ions, thereby propagating the potential from the axon hillock down the axon.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Sodium

Question **39**

Not answered

Marked out of 1.00

Flag question

Which of the following is a common biochemical finding in depressed patients?

Select one:

- Reduced Cortisol levels
- Raised ACTH levels
- Reduced Thyroid stimulating hormone
- Raised Corticotropin releasing hormone
- Raised Thyrotropin-releasing hormone

Your answer is incorrect.

Depression is associated with raised CRH (Corticotropin releasing hormone) and raised cortisol levels. A blunted ACTH level is also associated with depression.

The correct answer is: Raised Corticotropin releasing hormone

Question **40**

Not answered

Marked out of 1.00

Flag question

You are a junior doctor in liaison psychiatry. You have been asked to review Richard, a 25-year-old male who sustained severe head injury in a motorcycle race event. On recovery, he is noted to have a grasp reflex, poor judgement and impulsive reasoning. You believe his executive functioning has been adversely affected. Which of the following test would be most appropriate to confirm your suspicion?

Select one:

National adult reading test

- Paragraph retention test
- Weschler memory scale
- Weschler adult intelligence scale
- Wisconsin card sorting test

Your answer is incorrect.

The Wisconsin Card Sorting Test (WCST) contains stimulus cards of different colour, form and number. These are presented to patients to sort in groups according to single principle (e.g. to sort by colour, ignoring form and number). Individuals with damage to frontal lobes or to the caudate, as well as individuals with schizophrenia, give abnormal responses.

The correct answer is: Wisconsin card sorting test

Question **41**

Not answered

Marked out of 1.00

Flag question

Abnormalities in the metabolism of neurotransmitters are implicated in various mental health disorders. Which of the following neurotransmitters is produced by the action of glutamate decarboxylase?

Select one:

- Adrenaline
- Glutamate
- Dopamine
- GABA
- Sertraline

Your answer is incorrect.

Glutamate decarboxylase or glutamic acid decarboxylase (GAD) is an enzyme that catalyses the decarboxylation of glutamate to GABA and carbon dioxide.

The correct answer is: GABA

Question **42**

Not answered

Marked out of 1.00

Flag question

The stages of normal sleep can be distinguished by specific findings on EEGs. Which of the following stages of sleep is associated with 12-16Hz frequency waves on EEG?

Select one:

- Deep sleep
- Slow wave sleep
- Stage 2 sleep
- REM sleep
- Stage 1 sleep

Your answer is incorrect.

Stage 1 sleep is associated with a reduction of alpha wave rhythm. Stage 2 sleep is associated with sleep spindles, which have a frequency of 12-16Hz (usually 14Hz). Stage 3 and 4 are generally grouped together as slow-wave sleep. The distinction between stage 3 and stage 4 sleep is only a quantitative one that has to do with the amount of delta activity. Stage 3 is defined by delta activity that occupies 20-50% of the time, whereas in stage 4, delta activity represents greater than 50% of the time. REM sleep is associated with EEG desynchronization. The EEG background activity changes from that seen in slow wave sleep to faster and lower voltage activity (theta and beta), resembling wakefulness. Saw tooth waves are a special type of central theta activity that has a notched morphology resembling the blade of a saw and usually occurs close to rapid eye movements.

The correct answer is: Stage 2 sleep

Question **43**

Not answered

Marked out of 1.00

Flag question

During a neurological exam of a patient, you notice that the patient is unable to recognise a pound coin when you place it in their hand having asked them to close their eyes. Which of the following describes this phenomenon?

Select one:

- Anosognosia
- Simultagnosia
- Visual agnosia
- Prospognosia
- Astereognosis

Your answer is incorrect.

Astereognosis is the phenomenon where a person is unable to recognise an object with their hands when another sensory input (usually visual) is removed. Prosopagnosia is inability to recognise faces. A person is unaware of neurological impairment in Anosognosia. Visual agnosia is inability to recognise everyday objects. Simultagnosia is inability to process all aspects of visual input as a whole (i.e. only process each piece of furniture, people in the room etc separately).

The correct answer is: Astereognosis

Question **44**

Not answered

Marked out of 1.00

Flag question

Mr Xavier is an 85-year-old man who has experienced a stroke affecting his dominant parietal lobe. Which of the following signs is he most likely to demonstrate on examination?

Select one:

- Abulia
- Cortical deafness
- Finger agnosia
- Expressive aphasia
- Simultanagnosia

Your answer is incorrect.

Damage to the dominant parietal lobe can lead to Gerstmann syndrome, which is characterised by: dysgraphia, dyscalculia, finger agnosia and left-right disorientation.

The correct answer is: Finger agnosia

Question **45**

Not answered

Marked out of 1.00

Flag question

Which of the following describes rod shaped inclusion bodies seen in both normal ageing and Alzheimer's disease?

Select one:

- Tau protein
- TDP-43
- Amyloid proteins
- Lewy bodies
- Hirano bodies

Your answer is incorrect.

Hirano bodies are rod-shaped inclusion bodies seen in both normal aging and Alzheimer's disease. The other proteins listed are also associated with various forms of dementia.

The correct answer is: Hirano bodies

Question **46**

Not answered

Marked out of 1.00

Flag question

Mr. Young is a 78-year-old man with a diagnosis of vascular dementia. His family contacts the GP to raise concerns that he no longer appears to recognise people by sight, even those he knows well. Which is the following describes this phenomenon?

Select one:

- Prosopagnosia
- Abulia
- Astereognosis
- Anosognosia
- Akinetopsia

Your answer is incorrect.

Prosopagnosia refers to an inability to recognise people's faces. It is associated with damage or impairment of the fusiform gyrus. Astereognosis refers to the inability to identify an object by touch alone. Abulia refers to a profound lack of will and initiative. Anosognosia is the inability to recognise one's own neurological or physical impairments. Akinetopsia is the inability to perceive visual motion.

The correct answer is: Prosopagnosia

Question **47**

Not answered

What does the Stroop Test assess?

Select one:

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Flag question

- Working memory
- Problem solving
- Set shifting
- Organisation and planning
- Selective attention

Your answer is incorrect.

The Stroop test involves naming the colour of a word, which may or may not match with the word itself (if it's not a match, it is a different colour). It tests selective attention and inhibitory control.

The correct answer is: Selective attention

Question **48**

Not answered

Marked out of 1.00

Flag question

Different neurones utilise various neurotransmitters to convey signals in the brain. Which of the following is the most abundant fast excitatory neurotransmitter in the brain?

Select one:

- Glutamate
- Acetylcholine
- Glycine
- Serotonin
- Adrenaline

Your answer is incorrect.

Glutamate is the primary fast excitatory neurotransmitter in the brain.

The correct answer is: Glutamate

Question **49**

Not answered

Marked out of 1.00

Flag question

Billy Bailey is a 9-year-old boy with a diagnosis of ADHD. He was trialled on methylphenidate but experienced impaired growth. It is decided that he should now have a trial of atomoxetine. Which of the following receptors does it affect?

Select one:

- Dopamine
- Noradrenaline
- Serotonin
- Glutamate
- GABA

Your answer is incorrect.

Atomoxetine is a noradrenaline reuptake inhibitor licensed for the treatment of ADHD.

The correct answer is: Noradrenaline

Question **50**

Not answered

Marked out of 1.00

Flag question

The amphetamine challenge test involves administering a single dose of d-amphetamine to an individual and then performing functional scans to detect levels of glucose metabolism or dopamine release. In people with schizophrenia, which of the following areas shows increased dopaminergic release during this test?

Select one:

- Hippocampus
- Prefrontal cortex
- Periventicular areas
- Anterior temporal lobes
- Striatum

Your answer is incorrect.

People with schizophrenia demonstrate an increased release of dopamine in the striatum compared to controls on the amphetamine challenge test. Glucose metabolism studies have shown increased activity in the left temporal lobe.

The correct answer is: Striatum

Question **51**

Not answered

Marked out of 1.00

Flag question

In the neurobiology of alcohol addiction, from where is the initial release of dopamine?

Select one:

- Amygdala
- Ventral-tegmental area
- Hippocampus
- Periaqueductal area
- Nucleus Accumbens

Your answer is incorrect.

The ventral-tegmental area (VTA) is one of the components of the mesolimbic dopamine pathway. The neurones (cell bodies located in the VTA) stretch to the nucleus accumbens (NA). The VTA neurones release dopamine which acts on neurones in the NA, an area of the brain associated with motivation, reward and reinforcement learning. (Ref: Clapp, P., Bhave, S. V., & Hoffman, P. L. (2008). How Adaptation of the Brain to Alcohol Leads to Dependence: A Pharmacological Perspective. Alcohol Reasearch & Health, 31(4), 310-339).

The correct answer is: Ventral-tegmental area

Question **52**

Not answered

There are various histological findings that are associated with organic brain disorders. Which of the following histological features are described as being carrot-shaped eosinophilic intracellular

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Flag question

inclusions?

Select one:

- Hirano body
- Neurofibrillary tangle
- Lewy body
- Senile plaque
- Rosenthal fibres

Your answer is incorrect.

Rosenthal fibres are thick, elongated (resemble a sausage, or carrot or "corkscrew") eosinophilic bundle that can be found in the context of long-standing gliosis, low-grade gliomas, and some metabolic disorders. Lewy bodies are eosinophilic cytoplasmic inclusions consisting of a dense core surrounded by a halo of 10nm wide radiating fibrils. Hirano bodies are often described as rod-shaped, crystal-like, and eosinophilic and can be present in Alzheimer's disease and CJD.

The correct answer is: Rosenthal fibres

Question **53**

Not answered

Marked out of 1.00

Flag question

In a patient with schizophrenia, which pathway of the brain is thought to have excessive dopamine transmission

Select one:

- Pyramidal pathway
- Posterior column pathway
- Prefrontal pathway
- Cerebellar pathway
- Mesolimbic pathway

Your answer is incorrect.

Kapur proposed that in the normal individual, the role of mesolimbic dopamine is to attach significance or 'salience' to an external stimulus, or an internal thought. This converts a neutral piece of information into an attention-grabbing information (Kapur, 2003). In acute psychosis, where hyperdopaminergic state is noted in the mesolimbic system, insignificant events and perceptions receive inappropriate salience. For example, an innocuous smile of a stranger may be given high degree of 'aberrant salience' leading to delusional elaborations. On a similar note, when such aberrant salience is attached to internally generated self-speech, hallucinations may be experienced. Antipsychotics are claimed to 'dampen the salience' of these abnormal experiences (not erase them) but provide a platform for a process of psychological resolution.

The correct answer is: Mesolimbic pathway

Question **54**

Not answered

Marked out of

Mark Thurston came back from an army tour three months ago, where he was involved in front-line conflict and exposed to situations in which he witnessed others being subject to major injury. For the past six weeks, he's been experiencing PTSD symptoms, with particularly disturbed sleep. Which of

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Flag question

the following would benefit Mr. Thurston the most in relation to his PTSD and sleep disturbance?

Select one:

- Increasing the number of non-REM sleep cycles
- Reducing the amount of waking from REM sleep
- Increasing the length of stage 3 and 4 sleep
- Shortening REM latency
- Reducing jerks during sleep

Your answer is incorrect.

REM sleep disruption is implicated in PTSD, mediated by sympathetic system over-activity. REM sleep starts when the sympathetic nervous system and noradrenaline secretion are at their lowest level. Prazosin helps restore REM sleep cycles, which may help alleviate sleep disturbance and nightmares in PTSD.

The correct answer is: Shortening REM latency

Question **55**

Not answered

Marked out of 1.00

Flag question

Which of the following neurotransmitters is most abundant in the autonomic ganglia?

Select one:

- Serotonin
- Noradrenaline
- Dopamine
- Histamine
- Acetylcholine

Your answer is incorrect.

Autonomic ganglia are clusters of nerve cell bodies within the autonomic nervous system, of which there are two types: sympathetic and parasympathetic. Acetylcholine is the main neurotransmitter in the preganglionic sympathetic and parasympathetic neurons.

The correct answer is: Acetylcholine

Question **56**

Not answered

Marked out of 1.00

Flag question

Woczeck is a 34-year-old man with a diagnosis of schizophrenia. For some years he's been telling mental health professionals that his arms are under the control of the Prime Minister, and they can be raised and lowered of their own accord, without Woczeck being able to exert his influence over these movements. Control delusions such as these are related to what theory of information processing?

Select one:

- Working memory
- Analytic processing
- Sequential processing
- Parallel distributed processing

- Sensory prediction

Your answer is incorrect.

Motor control strongly relies on neural processes that predict sensory consequences of self-generated action. Deficits in this process are thought to be implicated in delusions of control.

The correct answer is: Sensory prediction

Question **57**

Not answered

Marked out of 1.00

Flag question

In which of the following conditions would EEG be most helpful diagnostically?

Select one:

- Huntington's disease
- Parkinson's disease
- Familial CJD
- Paranoid schizophrenia
- Pseudodementia

Your answer is incorrect.

In Huntington's disease, there is an initial loss of alpha wave activity and an eventual flattened trace. An EEG would reveal no typical abnormalities in any of the other conditions listed.

The correct answer is: Huntington's disease

Question **58**

Not answered

Marked out of 1.00

Flag question

The mother of a patient in your CAMHS clinic asks for clarification around genetic anticipation. With which of the following disorders is anticipation associated?

Select one:

- Fragile X syndrome
- Turner syndrome
- Angelman syndrome
- Prader-willi syndrome
- Down's syndrome

Your answer is incorrect.

Anticipation is the term given to the phenomenon that some genetic disorders tend to appear earlier in age, with more severe symptoms as the disorder is passed down from one generation to the next. It is associated with, amongst others, Fragile X and Huntington's chorea.

The correct answer is: Fragile X syndrome

Question **59**

Not answered

Marked out of 1.00

Flag question

Mr. Stevens is a 24-year-old man with a diagnosis of amphetamine misuse. Which of the following parts of the brain is most directly affected by amphetamine use?

Select one:

- Parietal lobe
- Striatum
- Temporal lobe
- Prefrontal cortex
- Corpus callosum

Your answer is incorrect.

Amphetamine use is associated with changes to the dopaminergic neurons in the striatum, particularly in the caudate and putamen.

The correct answer is: Striatum

Question **60**

Not answered

Marked out of 1.00

Flag question

Miss Taylor is a 25-year-old woman with a diagnosis of autism spectrum disorder. Which of the following neurological findings is associated with autism spectrum disorders?

Select one:

- Increased cingulate volume
- Decreased volume of caudate nucleus
- Decreased overall brain volume
- Decreased volume of corpus callosum
- Decreased amygdala volume

Your answer is incorrect.

Neuroimaging studies have demonstrated that people with a diagnosis of autism have, on average, a smaller volume corpus callosum than controls. They are also demonstrated that people with autism spectrum disorder have increased volume of the caudate nucleus and amygdala, as well as an overall larger brain size. They also tend to have a smaller cingulate gyrus.

The correct answer is: Decreased volume of corpus callosum

Question **61**

Not answered

Marked out of 1.00

Flag question

Different neurons utilise various neurotransmitters to convey signals in the brain. Which of the following neurotransmitters shuttle through astrocytes?

Select one:

- Serotonin
- Glycine
- Dopamine
- Glutamate

- GABA

Your answer is incorrect.

Astrocytes can release glutamate, ATP, nitric oxide and various prostaglandins.

The correct answer is: Glutamate

Question **62**

Not answered

Marked out of 1.00

Flag question

Mirtazapine is one of the most commonly used antidepressants in clinical practice. Which of the following most accurately describes its mechanism of action?

Select one:

- Noradrenergic and specific serotonergic antidepressant
- Noradrenaline reuptake inhibitor
- Serotonin and noradrenaline reuptake inhibitor
- Monoamine oxidase inhibitor
- Reversible inhibitor of monoamine oxidase A

Your answer is incorrect.

Mirtazapine is classified as a noradrenergic and specific serotonergic antidepressant.

The correct answer is: Noradrenergic and specific serotonergic antidepressant

Question **63**

Not answered

Marked out of 1.00

Flag question

Numerous substances can influence the metabolism of medications, sometimes resulting in impaired efficacy or toxicity. Which of the following substances are most likely to affect the function of oral contraceptives?

Select one:

- Tobacco
- St John's Wort
- Grapefruit juice
- Quetiapine
- Citalopram

Your answer is incorrect.

St John's Wort induces the enzymes CYP3A4 and CYP1A2 and increases the metabolism of the combined oral contraceptive pill, leading to an increased risk of contraceptive failure.

The correct answer is: St John's Wort

Question **64**

Which of the following neurotransmitters is most abundant at the locus coeruleus?

Not answered

Marked out of 1.00

Flag question

Select one:

- Dopamine
- Acetylcholine
- Noradrenaline
- Serotonin
- Glutamate

Your answer is incorrect.

The locus coeruleus is located in the pons and is part of the reticular activating system. It is the principal area for the synthesis of noradrenaline.

The correct answer is: Noradrenaline

Question **65**

Not answered

Marked out of 1.00

Flag question

Mrs. Baker is a 78-year-old woman who has recently been diagnosed with Alzheimer's disease. Her daughter informed her old age psychiatrist that 2 of Mrs. Baker's 4 siblings have also received the same diagnosis. She asks about the possible genetic factors involved in the development of Alzheimer's disease. How many times more likely is a patient to develop Alzheimer's disease if they are a heterozygote for ApoE4?

Select one:

- 4 times
- 6 times
- 10 times
- 1.5 times
- 2 times

Your answer is incorrect.

ApoE2 is protective against Alzheimer's disease. Most of the population (around 60%) have 'double dose' of the APOE3 gene and therefore are at 'average risk'. An individual is 3 to 4 times more likely to develop Alzheimer's disease if they are heterozygous for ApoE4 (i.e. one copy). Nearly 25% of the population is homozygous. This increases to 10-15 times if they are homozygous (i.e. 2 copies) for ApoE4.

The correct answer is: 4 times

Question **66**

Not answered

Marked out of 1.00

Flag question

You are a junior doctor in a learning disabilities team. Your next patient is eight-year-old Rupert who presents with moderate mental retardation. On observation, he has rather pronounced features, including a broad forehead and wide mouth. His past medical history includes aortic stenosis and his blood tests reveal hypercalcaemia. You suspect that he has William's syndrome. Which of the following tests would you use to confirm this diagnosis?

Select one:

- Western blot

- Southern blot
- FISH
- CISH
- Karyotyping

Your answer is incorrect.

William's syndrome is a rare genetic condition that results in a developmental disorder. It is characterized by mild to moderate mental retardation, unique personality characteristics, distinctive facial features, and cardiovascular problems. Williams' syndrome is caused by the microdeletion of 7q11 on chromosome 7. Clinically, the diagnosis can be confirmed by the FISH technique. This blood test, known as fluorescent in situ hybridization, detects elastin deletion on chromosome 7 and has a sensitivity of >98%.

The correct answer is: FISH

Question **67**

Not answered

Marked out of 1.00

Flag question

Which of the following describes the function of an endosome?

Select one:

- Involved in the degradation of lipids
- Involved in the production of hydrogen peroxide
- Involved in the release of energy
- Involved in protein synthesis
- Involved in the storage of materials

Your answer is incorrect.

Endosomes are involved in the degradation of lipids. LDL reaches early endosomes and dissociates from its receptor, remaining in the endosome before being delivered to the lysosome for degradation. The mitochondria is involved in releasing energy via aerobic respiration, a by-product of which is hydrogen peroxide. Protein synthesis occurs on the ribosomes and the vacuole is involved in storage.

The correct answer is: Involved in the degradation of lipids

Question **68**

Not answered

Marked out of 1.00

Flag question

Which of the following gene products is associated with Frontotemporal dementia?

Select one:

- Gamma-synuclein
- Alpha-synuclein
- Amyloid beta protein
- Tau protein
- Beta-synuclein

Your answer is incorrect.

Familial fronto-temporal dementia is associated with mutations in three genes: C9ORF72, tau protein (MAPT) and progranulin (PRG). Amyloid beta protein is associated with Alzhiemer's disease. Alpha and Beta-synuclein are found in Parkinson's disease. Alpha-synuclein is associated with Lewy body dementia. Gamma-synculein is found primarily in the paeripheral nervous system and is associated with breast tumour development.

The correct answer is: Tau protein

Question **69**

Not answered

Marked out of 1.00

Flag question

Miss Jameson is a 26-year-old woman with a diagnosis of schizophrenia. She is currently well and attending a follow-up clinic with you. Whilst there she enquires about the genetic factors influencing the development of schizophrenia. Which of the following genes is most associated with schizophrenia?

Select one:

- Dysbindin
- Corticotropin-releasing hormone receptor 1
- Huntington
- Dysregulin
- Progranulin

Your answer is incorrect.

Dysbindin and neuregulin are associated with schizophrenia. Corticotropin-releasing hormone is related to depression. Progranulin is associated with frontotemporal degeneration. Triple repeat expansion in Huntingtin is the cause of Huntington's disease.

The correct answer is: Dysbindin

Question **70**

Not answered

Marked out of 1.00

Flag question

The study of epigenetic mechanisms and their influence on the development of numerous heritable conditions is an expanding field of study, including within psychiatry. Which of the following epigenetic mechanisms is most associated with the development of schizophrenia?

Select one:

- DNA methylation
- Genomic imprinting
- Micro RNA modification
- Nucleosome positioning
- RNA transcripts

Your answer is incorrect.

DNA methylation, particularly of GABA-ergic genes and brain derived neurotrophic factor, are suspected to be linked to schizophrenia.

The correct answer is: DNA methylation

Question **71**

Not answered

Marked out of 1.00

Flag question

Which of the following describes the process of producing a protein from an RNA molecule?

Select one:

- Cell duplication
- Protein synthesis
- Mitosis
- Translation
- Transcription

Your answer is incorrect.

Translation is the process of producing a protein molecule from an RNA transcript. Transcription is the process of producing an RNA molecule from DNA.

The correct answer is: Translation

Question **72**

Not answered

Marked out of 1.00

Flag question

Heritability is a measure of:

Select one:

- Genetic contribution to phenotypic variation in a population
- Risk of disease in first degree relatives of probands
- Genetic contribution of disease in an individual
- Degree of penetrance of a disease
- Concordance of disease between twins

Your answer is incorrect.

The relative influence of genetic factors in determining the variance in a trait is expressed as heritability.

The correct answer is: Genetic contribution to phenotypic variation in a population

Question **73**

Not answered

Marked out of 1.00

Flag question

What percentage of the cerebrum is comprised of the frontal cortex?

Select one:

- 37-41%
- 15-20%
- 5-10%
- 20-25%
- 70-75%

Your answer is incorrect.

The frontal lobe is the largest lobe in the cerebrum, comprising of between 37 and 41%. The temporal lobe is 22-24% of the cerebrum, parietal lobe is 19-20% and the occipital is 18-19%.

The correct answer is: 37-41%

Question **74**

Not answered

Marked out of 1.00

Flag question

Joe is an 8-year boy whom you assess in your CAMHS clinic. You note that he lacks eye contact and appears interested in his own play. His mother reports that he does not play games with the other children at school, preferring more repetitive solo activities. Hypoplasia of which region of the brain is associated with this condition?

Select one:

- Hypothalamus
- Prefrontal cortex
- Hippocampal formication
- Cerebellar vermis
- Parieto-occiptal cortex

Your answer is incorrect.

The above description suggests the child has autism, which is associated with hypoplasia of the cerebellar vermis.

The correct answer is: Cerebellar vermis

Question **75**

Not answered

Marked out of 3.00

Tau protein
Ubiquitin
Amyloid A protein
Beta-amyloid protein
Alpha-synuclein protein
Iron granules
FUS protein

Neuropathology

The subtypes of dementia are associated with abnormal molecules accumulating in the cerebrum. Which of the following molecules are most likely be found in the below types of dementia?

1. Mrs. Gasker is a 72-year-old woman who has attended her GP as her family is increasingly worried about her ability to manage at home due to cognitive problems, although her memory appears to be intact. They have also noted that she has been experiencing tremor and limb stiffness. She has started to have falls in the last few weeks. (Please choose TWO answers)

2. Mrs. Hunter is a 76-year-old woman who has been exhibiting increasingly odd behaviour over the past 18 months. Her family describe her as being "sensitive" and "diligent" but she has started to make lewd jokes and indifferent to the concerns of others. Her husband has noted that she is finding it difficult to look after herself properly. Please choose ONE option.

Explanation:

1) The first case describes a history of dementia with Lewy bodies. The key constituent of Lewy

bodies is **alpha-synuclein.** When aggregates of misfolded, overexpressed alpha-synuclein accumulate in neurons (axonal processes and synapses) or surrounding glial cells, Lewy bodies are formed. Lewy bodies are immunoreactive for alpha-synuclein as well as for **ubiquitin** and alpha-beta-crystallin. Neuropathic changes of dementia with Lewy bodies include the presence of Lewy bodies and Lewy neurites in the cerebral cortex as well as in the brainstem. These are often accompanied by some degree of Alzheimer disease neuropathologic changes, often with a moderate beta-amyloid plaque burden but relatively lower tangle burden.

2) The second case describes an example of fronto-temporal dementia. This is associated with **tau protein** deposition, particularly in Pick's disease. Pick disease is characterised by the presence of Pick bodies: rounded, homogeneous neuronal cytoplasmic inclusions, faintly visible on H&E staining. Immunohistochemical staining shows that these inclusions contain 3Rtau. Pick bodies are also strongly argyrophilic with the Bielschowsky technique.

Ref: Sanford AM. Lewy body dementia. Clinics in geriatric medicine. 2018 Nov 1;34(4):603-15.

Gray F, Duyckaerts C, de Girolami U. Escourolle and Poirier's Manual of Basic Neuropathology , 6th Edition. 2019. pp. 6-7, 192-193, 197.

The correct answer is:

1) Mrs. Gasker - Alpha-synuclein protein, U biquitin

2) Mrs. Hunter - T au protein

Question **76**

Not answered

Marked out of 4.00

Neurohistology 2

Each of the cell types in the central and peripheral nervous systems have particular characteristics. Please match the following cell types to the descriptions below.

1. Which cell types commonly use glutamate as a neurotransmitter? (Please choose TWO answers)

2. Which cell type transmits motor signal from the cortex to the spinal cord?

3. Which cell type lines the cerebral ventricles?

Chandelier cells
Astrocytes
Double bouquet cells
Pyramidal cells
Schwann cells
Granular cells
Ependymal cells
Oligodendrocyte

Explanation:

1) Numerous neurons utilise glutamate as a neurotransmitter, including **granular** and **pyramidal** cells. Glutamate is also shuttled via astrocytes, though these cells do not use glutamate as a neurotransmitter primarily.

2) **Pyramidal** cells, particularly Betz cells, are part of the corticospinal tracts that conveys motor signals from the motor cortex to the spinal cord.

3) **Ependymal** cells line the cerebral ventricles.

Question **77**

Neuropharmacology

Medications used in psychiatric practice often cause side effects, either because of their effects on

Not answered

Marked out of 4.00

target receptors in different areas of the body or affinity to non-target receptors. Which of the following receptors are associated with the below side-effects?

1. Mrs Oliver is a 32-year-old woman taking amisulpride who attends clinic and informs you that her periods have stopped. [ ]

2. Mr Pierce is a 52-year-old man who has started a new medication but describes episodes of dizziness in the mornings or when standing up. [ ]

3. Mr Radford is a 68-year-old man who has started taking an antidepressant he had many years ago. He attends his GP reporting difficulty-voiding urine. [ ]

4. Mrs Stevens is a 45-year-old woman who has been on a medication for 6 weeks but is keen to change to an alternative as she finds she is sleeping excessively and finds it difficult to wake in the mornings. [ ]

Histamine receptors
Alpha-1-adrenergic receptors
D2 receptors
Glutamate receptors
Muscarinic receptors
Nicotinic receptors
μ receptors
5-HT2A

Explanation:

1) Most antipsychotics **block D2 receptors** and have the potential to cause hyperprolactinaemia due to blockade in the pituitary. This is more frequent in patients taking amisulpride, risperidone, paliperidone or a first generation antipsychotic. Common consequences of this are infertility and osteoporosis in the long-term. Galactorrhoea is also possible in both men and women.

2) The second scenario appears to describe episodes of postural hypotension which can be linked to antagonism of **alpha-1-adrenergic** receptors. TCAs, mirtazapine and some antipsychotics are associated with this effect.

3) **Antimuscarinic** effects, which are common side-effects of TCAs, can cause urinary retention, as well as dry mouth, blurred vision and a variety of other effects.

4) **Histamine antagonism**, such as that caused by mirtazapine, can cause marked sedation, which some patients find undesirable. Promethiazine and quetiapine also have this effect.

Question **78**

Not answered

Marked out of 4.00

Function of Neuroglial cells
Identify the Neuroglial cells with their functions;

1. Cells which provide myelination to central nervous system [ ]

2. Cells which provide myelination to peripheral nervous system [ ]

3. Cells which transport glutamate to promote clearance from synapse [ ]

Astrocyte
Stellate cells
Chandelier cells
Oligodendrocytes
Schwann cells
Ependymal cells
Double bouquet cells

4. Cells which line the ventricles of the brain and the central canal of spinal cord

Explanation:

Oligodendrocytes provides myelination to central nervous system and **Schwann** cells provides myelination to peripheral nervous system.

Rapid removal of glutamate from the synaptic space is required for the survival of releasing neurons. Although glutamate transporters are expressed by all CNS cell types, **astrocytes** are the cell type primarily responsible for glutamate uptake.

Ependymal cells line the CSF filled ventricles in the brain and the central canal of spinal cord.

Ref: https://www.ncbi.nlm.nih.gov/pubmed/10975906

Question **79**

Not answered

Marked out of 5.00

Switching antidepressants
What is the advised route for the following changes in medication?

Phenelzine to Sertraline — Choose...

Fluoxetine to duloxetine — Choose...

Fluoxetine to Tranylcypromine — Choose...

Trazodone to Mirtazapine — Choose...

Amitryptiline to citalopram — Choose...

Choose...
Withdraw and wait for 2 weeks
Withdraw and wait for 5-6 weeks
Stop and start other drug immediately
Half dose, start new drug, then slow withdrawal
Concomitant prescription is permissible
Withdraw and wait for 1 week
Cross taper with caution

Your answer is incorrect.

When swapping from one antidepressant to another, abrupt withdrawal should usually be avoided. Cross tapering is preferred, where the dose of the ineffective or poorly tolerated drug is slowly reduced while the new drug is slowly introduced. 1) Cross tape

The correct answer is: Phenelzine to Sertraline → Withdraw and wait for 2 weeks, Fluoxetine to duloxetine → Stop and start other drug immediately, Fluoxetine to Tranylcypromine → Withdraw and wait for 5-6 weeks, Trazodone to Mirtazapine → Cross taper with caution, Amitryptiline to citalopram → Half dose, start new drug, then slow withdrawal

Question **80**

Not answered

Marked out of 8.00

QTc prolongation
Mr Barnes is a 78-year-old man with a diagnosis of schizophrenia and moderate depression. He has been trialled on a number of medications, some of them with only limited success. You are reviewing his notes and deciding on the next possible pharmacotherapeutic options. You are aware he has a family history of cardiac dysfunction, with ischaemic heart disease on the maternal side, and cardiac arrhythmia on the paternal side. Choose from the list of medications given, that has:

1. No effect on the QTc at therapeutic doses (choose 2 options)

IV Haloperidol
Venlafaxine
Citalopram
Aripiprazole
Mirtazapine
Risperidone
Sertindole
Quetiapine

2. Low effect on the QTc at therapeutic doses (choose 2 options)

3. Moderate effect on the QTc at therapeutic doses (choose 2 options)

4. High effect on the QTc interval at therapeutic doses (choose 2 options)

1) **No effect** on the QTc at therapeutic doses- Drugs include **Aripiprazole** and **Mirtazapine**
2) **Low effect** on the QTc at therapeutic doses- Drugs include **Venlafaxine** and **Risperidone**.
3) **Moderate effect** on the QTc at therapeutic doses- Drugs include **Citalopram** and **Quetiapine**.
4) **High effect** on the QTc at therapeutic doses- Drugs include **Haloperidol**, **Sertindole** and **any intravenous antipsychotic.**

(Ref: The Maudsley Prescribing guidelines- 11th edition- page 117)

Question **81**

Not answered

Marked out of 3.00

Akathisia
Parkinsonism
Neuroleptic malignant syndrome
Agranulocytosis
Tardive dyskinesia
Dystonia
Serotonin syndrome
Seizures

Complications of neuroleptics
In each of the below scenarios, which of the following options would be the most likely diagnosis?

1. Jamie is a 23-year-old man who was admitted following unusual and aggressive behaviour. He was experiencing auditory hallucinations and paranoid thoughts. He was started on medication and soon complained that his arm and neck muscles were rigid and twisted in a painful way.

2. Paula is a 68-year-old woman with a long history of schizophrenia. She attends your outpatient clinic complaining of strange involuntary movements in her face, particularly puckering and smacking her lips.

3. As the on call psychiatrist, you are called to review a newly admitted 25-year-old woman on the ward. The nurses state that she had been given IM haloperidol, and has since become unwell. On examination, you note that she appears confused and sweating, with rigid muscles.

Explanation: 1) Jamie is experiencing dystonia, which is characterised by rigidity and involuntary contraction of muscles following administration of a neuroleptic
2) Paula is experiencing tardive dyskinesia, an adverse effect of long term use of antipsychotic

medication 3) The patient has neuroleptic malignant syndrome , a rare, life-threatening adverse effect of antipsychotic use. The patient will exhibit symptoms including a high temperature (>38 degrees), autonomic instability, confusion or altered consciousness, rigid muscles and tremors.

Finish review

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